

DBCLS BioHackathon 2026 report: Open, Reproducible Extraction of Data-Entity References from the Biomedical Literature – with Entry-Level Resolution via Identifiers.org / TogoID

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Vijay Venkatesh Subramoniam ¹, **Yasunori Yamamoto** ², **Takatomo Fujisawa** ², and **Evan Bolton** ³

1 EMBL-EBI **2** BioData Science Initiative (BSI), NIG **3** National Center for Biotechnology Information, USA

Abstract

Funders and data producers need to know not just which databases are cited in the literature, but whether a cited accession actually exists and how it was used. Making that reuse visible is itself an incentive for the wider research community to deposit data, by giving concrete, attributable evidence that public deposits are genuinely reused and contribute to science. We report on a pipeline, built during the DBCLS BioHackathon 2026, that extracts database identifiers from PMC full text, verifies each one against source-of-truth APIs (TogoID/RDF Portal, NCBI E-utilities, EBI Search), and classifies confirmed accessions into one of five credit/provenance classes. On a working slice of 10,896 PMC documents, the pipeline confirms 19,786 (paper, accession) links across 1,352 papers and 18,159 distinct accessions, though a single namespace (*insdc*) accounts for roughly 70% of that volume. An independent comparison against Europe PMC's own accession annotations over the same slice shows 74.6% agreement at the individual-citation level, with a bimodal split at the per-document level: 64% of documents that either system found anything in agree at 90-100%, while 16% agree at only 0-10%, a pattern that needs manual citation and analysis. The pipeline, its Europe PMC comparison, and a static, backend-free dashboard reporting the results are complete and reproducible for this slice. A reference-role (generated/used/mentioned) classifier and a 221-item human gold standard evaluating it were also completed, reaching 0.80 macro-F1 with a near-zero generated-vs-used confusion; extending coverage and running against the full corpus are left as future work.

Introduction

Data producers and funding agencies increasingly need to know how deposited datasets are actually reused in the published literature, and to know it at the level of individual database entries (accessions), not merely which databases are cited. Beyond those stakeholders, this visibility speaks to the research community as a whole. A persistent barrier to data sharing is that what happens to a dataset after deposition is hard for depositors to trace, and hard to get credit for, which weakens the incentive to deposit at all. By making entry-level reuse concrete, verifiable, and attributable, that is, showing that a specific deposited accession was later reused and contributed to published science, the same infrastructure that lets funders track reuse also gives researchers a tangible reason to deposit in the first place. Demonstrating that public deposition pays off in downstream science is, in itself, one of the strongest ways to encourage more of it. This project builds an open, reproducible, regularly updated pipeline

that extracts references to data entities from the full text of PMC Open Access articles (~8 million article versions), verifies that each referenced identifier really exists, and normalizes references at the entry level. The hackathon goal is create this pipeline, harden the extraction and the identification layer, have a dashboard to showcase this information and also make a comparison between the existing EuropePMC Annotations API and our pipeline and see if there is any gaps we have uncovered.

Background and motivation

Europe PMC text-mines accession numbers from over 40 life-science databases and publishes them weekly on its FTP/bulk-download site (Europe PMC, 2024b), which is valuable and complementary to this work. Since that publication, coverage has increased to currently ~54 data resources. The programmatic form of these annotations is available through the Europe PMC Annotations API (Europe PMC, 2024a). The corpus and access method this pipeline consumes roughly 8 million PMC article versions on the `pmc-oa-opendata` S3 bucket and are documented by NCBI (NCBI, 2026).

What is still missing across existing efforts is a combination of:

- (1) a fully reproducible, openly specified extraction method
- (2) entry-level normalization across databases (cross-referencing an accession to a single canonical identity)
- (3) coverage of databases beyond the common deposition archives (model-organism, ortholog, and other reference resources)
- (4) a distinction between reference *roles* – data used, generated, or merely mentioned (now implemented; Results §8)

Together, these make reuse not just measurable but attributable — the property that turns a reuse metric into a deposition incentive. Identifiers.org is central to points (1)-(2): its registry (prefixes, patterns) and resolver define the canonical namespaces that make entry-level identity possible. Our existence-verification layer already relies on Identifiers.org-style URIs.

Methods

This section describes what we built during the hackathon: the extraction-to-dashboard pipeline itself (Figure 1), how it is compared against Europe PMC's own annotations, and the dashboard used to report the result.

Pipeline architecture

Figure 1 shows the data flow in five stages: full-text extraction, candidate/ mention deduplication, existence verification, credit classification, and reporting.

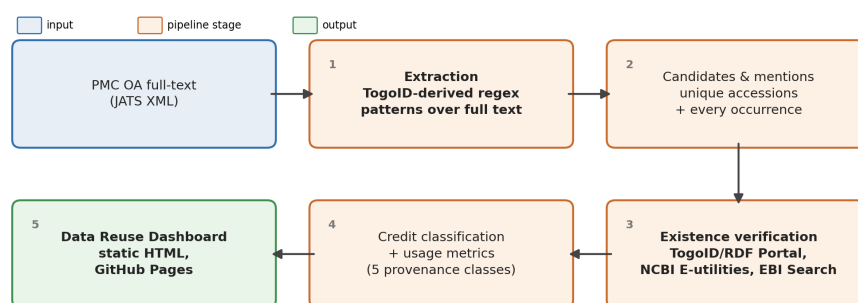


Figure 1: Pipeline architecture: PMC full text to a verified, entry-level data-reuse dashboard, in five stages.

Upstream inputs feeding this pipeline:

- **PMC OA full-text XML** addressed by shard via `shard_plan.json` (generated with `phase1_ner plan`).
- `togoid_dataset.yaml` – upstream DBCLS input, fetched from `dbcls/togoid-config -> config/dataset.yaml`.
- **rdf-config repository** (`git clone https://github.com/dbcls/rdf-config`) – input for the T2a registry.
- **Europe PMC Annotations API** / bulk dump, used as an independent comparison source, not as a dependency of extraction itself.

Pipeline stages. The pipeline proceeds through five stages.

Stage 1 (extraction) parses each JATS document with a shared full-text parser and applies the Togoid-derived regex patterns (Ikeda et al., 2022) (word-boundary anchored, digit-floor gated) to passage text, producing per-shard tables of raw accession mentions with their database, surface form, and text offset. No GPU is required, and each shard is processed idempotently.

Stage 2 (candidates and mentions) deduplicates raw mentions into a unique (database, surface) candidate set to be verified, while separately retaining every individual occurrence (with document ID and offset) for downstream counting.

Stage 3 (existence verification, T2 then T3) checks each candidate against source-of-truth registries in two tiers. T2 checks the Togoid label graph and `rdf-config`-native URIs against RDF Portal via SPARQL. T3 then confirms or refutes against two authoritative source APIs, routed by accession shape: NCBI E-utilities (Sayers, 2010) (authoritative for RefSeq/GEO) and EBI Search (Pearce et al., 2025). Each candidate ends this stage as confirmed, pending, or absent.

Stage 4 (metrics) joins confirmed entries with their mentions and classifies each into one of five credit/provenance classes (Table 1).

Table 1: Credit/provenance classes assigned to each confirmed identifier.

Class	Meaning
<code>deposited_research_record</code>	Primary research output deposited by the citing paper's own authors
<code>deposited_entity_registry</code>	Entity deposited in a registry as part of the described work
<code>curated_entity_registry</code>	Reference to a pre-existing, professionally curated registry entry
<code>derived_curated</code>	Reference to a derived/computed resource built on curated data

Class	Meaning
out_of_scope	Identifier-shaped string that does not represent an in-scope database citation

These five classes aren't from a single existing standard; we put them together ourselves, but not out of thin air. Two of the boundaries follow a distinction bioinformatics has used for decades: some databases just hold data as researchers submitted it (GenBank, GEO, SRA), while others take that raw data and curate or compute something new from it (UniProt, RefSeq, Pfam) (EMBL-EBI, [n.d.](#); Galperin et al., [2015](#)). That maps onto our `deposited_research_record/deposited_entity_registry` classes on one side and `derived_curated` on the other.

`curated_entity_registry` (HGNC, ChEBI, OMIM) doesn't fit either side of that split, so we added it as a third bucket. These aren't raw deposits, and nobody computed them from other data either, a curator looked at the evidence and decided "this gene is called BRCA1" or "this molecule is ChEBI 12345." That's a judgment call about identity, not a deposit or a derivation.

Stage 5 (reporting) turns the resulting usage tables into the dashboard described under Dashboard below.

This path is entirely separate from the concept/named-entity-recognition layer elsewhere in the repository (a GPU-backed HunFlair2 pipeline for biomedical concepts such as genes, diseases, organisms, and chemicals); the two share only the JATS full-text parser, and the path above needs no GPU.

Verification (T2/T3). Every candidate identifier is checked against a live registry before being counted as confirmed. T3 verification was extended during the hackathon to route by accession shape across **two** independent source APIs rather than one: NCBI E-utilities for RefSeq and GEO accessions, and **EBI Search** for INSDC/ENA nucleotide accessions. Specifically, nucleotide- and protein-style INSDC accessions must be routed to different services, protein accessions are not indexed under the `embl` domain EBI Search uses for nucleotide records, so protein-shaped IDs are instead verified through NCBI's `efetch` protein endpoint, while nucleotide-shaped IDs are confirmed against EBI Search directly. Adding EBI Search as its own verification layer closed a gap where nucleotide accessions had previously been under-verified through a domain mismatch.

Which route actually confirmed each link, on this run, is reported in Results §4 – read directly from this run's own dashboard data rather than from a separate benchmark.

Comparison with Europe PMC

We independently compare pipeline output against Europe PMC's own accession annotations for the same papers, rather than treating either pipeline's output as ground truth.

Comparison method. Europe PMC's accession annotations are fetched and normalized, mapping `identifiers.org` namespaces to the pipeline's own dataset keys and aligning annotation offsets to the local PMC XML text and then compared against this pipeline's own extraction output for the same documents. The comparison is over unique (PMC article ID, dataset, literal accession) triples, repeated mentions of the same accession within one article are not double-counted. Each triple is tagged both, `togoid_only`, or `europemc_only`, with database-level totals rolled up separately. These outputs are also rendered into an Excel workbook with per-group worksheets (Summary, Both, Europe PMC, TogoID, Comparison, TogoID only, Europe PMC only, Rejected review, Interpretation) and charts, for reviewers who prefer a spreadsheet over raw TSVs.

Import-quality checks are part of the normal run: rejected annotation-alignments are grouped by reason (an empty set on a fully aligned sample is the target), namespace-mapping ambiguities are surfaced separately, and DOIs and other non-accession identifiers are intentionally excluded from accession comparison rather than silently dropped.

Complementarity, not overlap.

Table 2: Europe PMC vs. this pipeline, on the axes that distinguish them.

Axis	Europe PMC	This pipeline
Core task	Text-mines full text for accessions + concepts (genes, diseases, organisms, chemicals)	Extracts accessions, then adds existence verification and role classification
Entry existence	Not verified – an annotation says “this string looks like a GEO ID,” not “this GEO record exists”	Verified against source (T2 TogoID/RDF Portal SPARQL; T3 NCBI E-utilities + EBI Search) -> confirmed/pending/absent
Reference role	Occurrence tags only – no notion of who did what	Classifies role: created (deposited) / used (reused) / mentioned
Database coverage	~54 databases, weekly, public API/FTP	~118-119 via TogoID, entry-type-split (RefSeq/UniProt/SRA/GEO broken into sub-types)
Method/output openness	Fully public (Kafkas method; Annotations API; weekly dump)	Ingests EPMC’s dump as a baseline input; layers verification + role classification on top

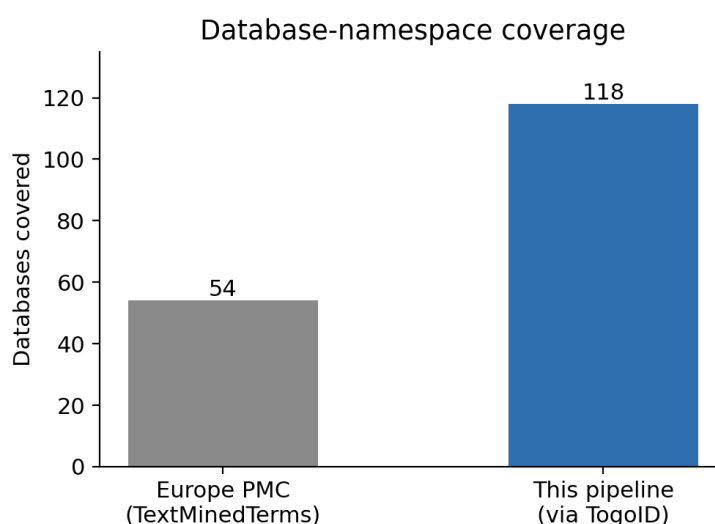


Figure 2: Database-namespace coverage: Europe PMC’s TextMinedTerms vs. this pipeline’s TogoID-derived pattern set.

Where each covers the other’s blind spot is the key finding (Figure 2): **EPMC covers what we currently miss** – our biggest recall gap is exactly the EBI deposition databases (ENA,

PDB, PRIDE, MetaboLights, EMPIAR, EMDDB, BioStudies, EGA, ArrayExpress, AlphaFold), which EPMC extracts well. **We cover what EPMC misses** – model-organism gene resources (FlyBase, MGI, RGD, SGD, WormBase, ZFIN, TAIR), orthology (COG, HomoloGene, OMA), and Japan-origin resources (JGA, NBDC, MBGD, TogoVar, GEA, NANDO). Downstream, EPMC’s text-mined accessions already feed the DataCite / Make Data Count Data Citation Corpus (its v4 ingested ~5.2M EPMC citations); this project sits above that aggregation layer, contributing verified, role-typed entries rather than raw mentions. In one line: EPMC answers “was this ID mentioned?”; this project answers “does the referenced dataset exist, and was it created or reused?” – the two coverage sets are complementary, not competing.

Dashboard

The “create a webpage to showcase the data” hackathon task was completed as a static, backend-free dashboard: a single self-contained HTML file with all data embedded inline and all filtering computed client-side. On our current working snapshot it reports **1,352 papers**, **18,159 distinct accessions** (curie-normalized, versions collapsed), **19,786 (paper, accession) links**, and **14.6 links per paper**, after excluding **291** out-of-scope links from the reporting denominator. It is deployed as a static site via GitHub Pages, requiring no backend infrastructure to serve to collaborators.

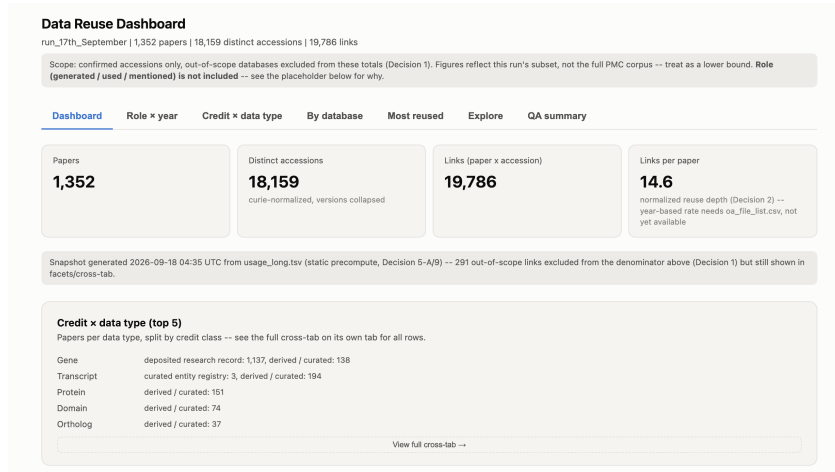


Figure 3: Data Reuse Dashboard: database usage drill-down

Reference-role classification

Credit/provenance *class* (Table 1) is a property of the *database* a confirmed accession belongs to. It does not say how a given paper *used* that accession. We therefore add an orthogonal, per-(paper, accession) axis — the reference **role** — distinguishing whether the citing paper **generated** the data (produced and deposited it), **used** it (reused a pre-existing dataset or reference resource as input), or merely **mentioned** it (named it as an entity, example, search hit, or tool/DB description). A used reference is further sub-tagged *ref_resource* when the thing used is a reference resource (e.g. aligning to a RefSeq/reference genome, querying UniProt) rather than reuse of a deposited dataset, so the two can be counted separately.

Two-stage classifier. For each confirmed link we build a context window around the marked accession occurrence by re-parsing the source JATS with the same full-text parser used in Stage 1, which reproduces byte-identical offsets so spans stay aligned. A rule pre-pass then routes by credit class: *out_of_scope* is skipped; *deposited_research_record*/*deposited_entity_registry* are sent in full to a language model (the rules contribute only an advisory prior); and *derived_curated*/*curated_entity_registry* are decided by high-precision

lexical rules (domain/family registries and un-cued mentions → mentioned; explicit reference-resource cues, e.g. “aligned to ... genome”, “as the query” → used), with ambiguous cases escalated to the model. The model stage runs a locally served open-weight instruction model (Qwen3.6-27B, FP8) under vLLM, data-parallel on 2×L40S GPUs, in non-thinking mode with schema-constrained (“guided”) JSON output; an Azure OpenAI backend is supported as a drop-in fallback. Each link receives a role, a short evidence span quoted from the window, and a confidence. On the role-classification slice this routes 41,341 links to rules and 12,135 to the model (missing/parse-error rate 0.4%).

Gold-standard construction. To evaluate the role axis we built a human gold standard of 221 (paper, accession) links, sampled from the confirmed set by stratified sampling over (rule/model source × credit class × predicted role), with a minimum floor per stratum and a cap on the dominant mentioned stratum so the credit-critical generated/used cells are adequately populated. Labelling was **blind to the model output**: the annotator saw only the accession, its context window, database, and section, and assigned one of generated, used, mentioned, or unsure (plus the `ref_resource` sub-tag), following the same definitions the classifier is given.

Results

Every number below is read directly from a sample run’s own output files – the Data Reuse Dashboard’s data (`metrics/usage_long.tsv`, built by `build_data_reuse_dashboard.py`) and the Europe PMC comparison matrix (`comparison/summary.json`, `comparison/by_database.tsv`, `comparison/document_agreement_histogram.tsv`, built by `compare_extra_actions.py`). The run covers 10,896 planned documents ; this is one working slice of PMC, not the full corpus, so nothing below should be read as a corpus-wide estimate.

1. Pipeline output volume

After the confirmed/in-scope denominator (Decision 1): **1,352 papers** have at least one confirmed accession, spanning **18,159 distinct accessions** (curie-normalized, versions collapsed) and **19,786 (paper, accession) links** – 14.6 links per paper on average. A further 291 links were confirmed but classified `out_of_scope` and excluded from these totals (still visible in the dashboard’s facets).

2. Credit/provenance class breakdown

Table 3: Confirmed links by credit/provenance class, this run.

Class	Papers	Accessions	Links
<code>deposited_research_record</code>	1,164	13,121	13,950
<code>derived_curated</code>	484	5,024	5,822
<code>curated_entity_registry</code>	6	14	14
<code>out_of_scope</code> (excluded above)	26	212	291

`deposited_research_record` dominates – about 70% of all links – almost entirely because of one database (below). `curated_entity_registry` is small in this run because few of the enabled namespaces fall in that class (see the credit-class discussion under Methods).

3. Data type and database concentration

By data type, **Gene** accounts for 1,193 of 1,352 papers and 14,974 of 19,786 links (~76% of all links); every other data type is a distant second (Transcript: 196 papers/2,258 links; Ortholog:

37 papers/858 links; Protein: 151 papers/820 links). Within Gene, a single database drives almost all of it: `insdc` alone accounts for 1,137 papers and 13,768 links – about **70% of every link in the dataset** comes from one namespace. The next-largest databases are `refseq_rna` (187 papers, 2,095 links), `refseq_protein` (119 papers, 731 links), `refseq_genomic` (103 papers, 934 links), `pfam` (51 papers, 437 links), and `cog` (37 papers, 858 links). Any claim about “database diversity” in this run has to be read against that concentration – the long tail is real, but it is a small fraction of total volume.

4. Verification route and confidence mix

Across all 20,077 confirmed links (before the out-of-scope exclusion in §1), the verification route that confirmed each one splits: EBI Search (`ena_api`) 11,813 (58.8%), Togoid label graph (`togoid_label`) 3,161 (15.7%), NCBI E-utilities (`ncbi_eutils`) 2,366 (11.8%), NCBI E-utilities protein route (`ncbi_eutils_protein`) 1,511 (7.5%), rdf-config-native RDF (`rdportal_native`) 771 (3.8%), and the Identifiers.org resolver (`idorg_resolution`) 455 (2.3%). Extraction confidence is overwhelmingly high: 20,022 links (99.7%) vs. 55 context-tier links (0.3%).

5. Most-reused accessions

The top-ranked “most reused” entries are `insdc:A23187` (29 papers) and `insdc:LY294002` (26 papers). The dashboard itself flags both as likely namespace collisions – A23187 and LY294002 are common compound/reagent codes that also happen to be syntactically valid ENA accessions – so this specific ranking needs manual review before being read as a clean citation-reuse signal.

6. Where accessions are cited (section distribution)

Of the 20,077 total confirmed links (before the out-of-scope exclusion), most fall in **results** (593 papers, 7,355 links) and **methods** (517 papers, 2,905 links) sections. **267 papers / 7,522 links (about 37% of all links)** could not be matched to a known section by the evidence-alignment step and are recorded as (`unknown`) – a real limitation of the current alignment logic, not a hidden gap: it is visible in the dashboard’s own section facet.

7. Comparison with Europe PMC

Across the same 10,896-document plan, the comparison matrix reports **15,546** distinct (paper, database, accession) triples found by **both** Togoid and Europe PMC, **4,476** found by Togoid only, and **807** found by Europe PMC only. Of the 20,829 total triples found by either system, that is **74.6% agreement**, **21.5% Togoid-only**, and **3.9% Europe-PMC-only**.

Per-document agreement is bimodal rather than evenly spread. Of the **1,370 documents** where either system found anything (12.6% of the 10,896 planned documents), **877 (64.0%) show 90-100% agreement** between the two systems, while **224 (16.4%) show 0-10% agreement** – near-total disagreement. This split is a real, currently unexplained pattern in the data, not an accuracy claim in either direction; the likely candidates (different namespace coverage between the two systems, differing normalization, or alignment/offset mismatches) have not yet been investigated per-document.

8. Reference-role classification: gold-standard evaluation

This evaluation is over the project’s role-classification slice (the feasibility subset, ~17.5k papers, 53,476 role-assigned confirmed links), not the 10,896-document verification run of §1–7; the two slices are being aligned. Of the 221 gold items, 18 labelled unsure are excluded, leaving **203** scored links.

Overall agreement between model and gold is **76.8%** (156/203), macro-F1 **0.80**. The credit-critical distinction is strong: **generated↔used confusions number 1**, and generated reaches F1 0.91. The entire error budget is one direction — used under-called as mentioned (Table 4).

Table 4: Role confusion matrix, model vs. gold (rows = gold, columns = model prediction).

gold model	generated	used	mentioned	total
generated	30	0	5	35
used	1	63	37	101
mentioned	0	4	63	67
total	31	67	105	203

Table 5: Per-role precision/recall/F1.

Role	Precision	Recall	F1	Support
generated	0.97	0.86	0.91	35
used	0.94	0.62	0.75	101
mentioned	0.60	0.94	0.73	67
macro	—	—	0.80	203

The error is concentrated and diagnosable. Split by source, the model stage is far stronger than the rule pre-pass (83.5%, 96/115 vs. 68.2%, 60/88). By credit class the deposited_research_record class — the one that matters for a data-reuse metric — scores **94%**, while derived_curated scores 68% and drives the average down. The mechanism is specific: of the 101 gold used links, 75 are reference-resource uses (ref_resource), and the rule pre-pass defaults reference-resource uses it cannot cue to mentioned; 19 of the 37 used→mentioned errors sit exactly in that rule path. Model confidence is well-calibrated for triage: links the model reports at confidence 1.0 are 95% correct, versus 70% at 0.9–0.99, so a confidence gate that routes sub-1.0 links to review isolates most errors.

Improvement identified. Because the missed used links are overwhelmingly the rule pre-pass defaulting reference-resource uses to mentioned, routing the derived_curated “default-mentioned” cases to the model instead (rather than finalizing them by rule) is projected to raise overall accuracy to ~82% (recovering ~12 of the 19 targeted used misses at the cost of ~10% of newly-seen true-mentioned links). This is a projection from the model’s observed per-label accuracy on derived_curated, not a re-measured run, and it trades cost (it pushes the large derived_curated mentioned volume onto the model); a measured re-run is left as an immediate next step.

Two caveats bound these numbers. The gold is drawn from the older-paper feasibility subset, and per-class supports are modest (curated_entity_registry $n \approx 28$), so per-class figures are directional and the reliable headline is the deposit-class result and the near-zero generated↔used confusion. And the evaluation scores the classifier as configured at sampling time; the routing change above requires re-running the model on the affected links before its effect is measured rather than projected.

Discussion

Several limitations are worth surfacing explicitly rather than folding into aggregate numbers.

One database dominates this run’s volume. insdc alone accounts for ~70% of every confirmed link (Results §3), driven almost entirely by the Gene data type. Any statement about

“database diversity” or “coverage” from this run has to be read against that concentration, the long tail of smaller databases (RefSeq variants, Pfam, COG, and the rest) is real, but it is a small fraction of total volume, and a corpus-wide run could look proportionally different depending on which slice of PMC it draws from.

A specific namespace-collision problem sits at the top of the “most reused” ranking. The two most-reused accessions in this run, `insdc:A23187` and `insdc:LY294002`, are also common compound/reagent codes that happen to be syntactically valid ENA accessions (Results §5). We have not independently re-verified these two entries in this pass; the dashboard surfaces the caveat visibly rather than silently filtering them, since there is no context-free way to tell “real citation” from “coincidental match” for this class of collision without closer manual review.

About a third of confirmed links have no resolved section. 7,522 of the links in this run (~37%) fall into (unknown) in the section breakdown (Results §6) because the evidence-alignment step could not match them to a known JATS section. That is a real gap in the current alignment logic, not a data-quality issue with the underlying accessions themselves, and it means any section-level claim (“most citations happen in Methods”) should be read as a claim about the ~63% of links that *did* align, not the full set.

Agreement with Europe PMC is bimodal. 64.0% of documents that either system found anything in show 90-100% agreement, but 16.4% show 0-10% agreement (Results §7), there is very little middle ground. We have not yet broken this down by database or by document to see whether the low-agreement group is concentrated in specific namespaces (which would point to a coverage gap on one side) or spread evenly (which would point to an alignment/normalization issue). That investigation is the natural next step before treating either system’s count as more reliable than the other’s.

This is one working slice of PMC, not the full corpus. The run covers 10,896 planned documents; none of the figures above should be extrapolated to the full PMC OA corpus without running the same pipeline over a larger or differently-sampled slice first.

Task status. The dashboard (Results §1-6) and the Europe PMC comparison (Results §7) are both complete and reproducible for this run. Reference-role (generated/used/mentioned) classification is now implemented and evaluated against a human gold standard (Methods; Results §8) on the role-classification slice; aligning it to the same document slice as §1–7 and a measured re-run of the routing improvement are the immediate next steps. Benchmarking against the Identifiers.org resolver directly and extending database coverage remain future work.

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